1 - Weight Loss Study

MATH 213

Group 1

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

All the participants in the study, healthy adults, aged 30 to 70, with BMI between 30 and 40

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: Treatment Group

Type: Categorical

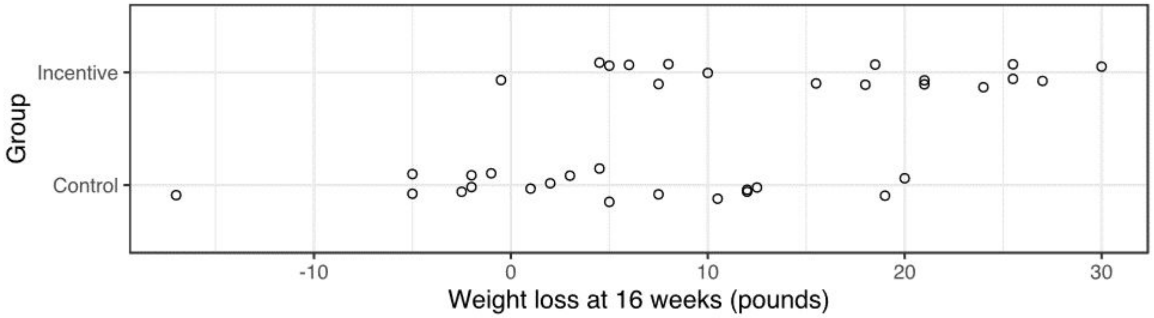
Response: Amount of Weight Lost

Type: Numerical

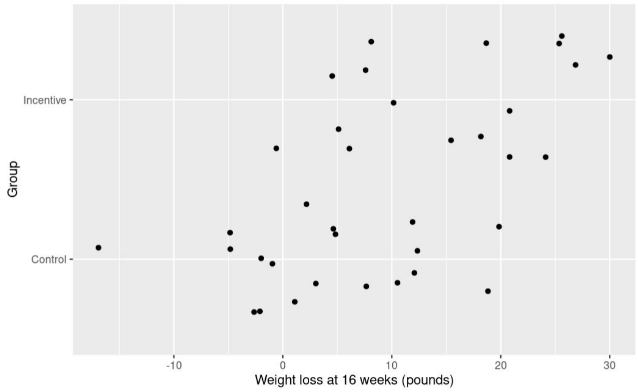
What makes this study an experiment? Explain briefly.

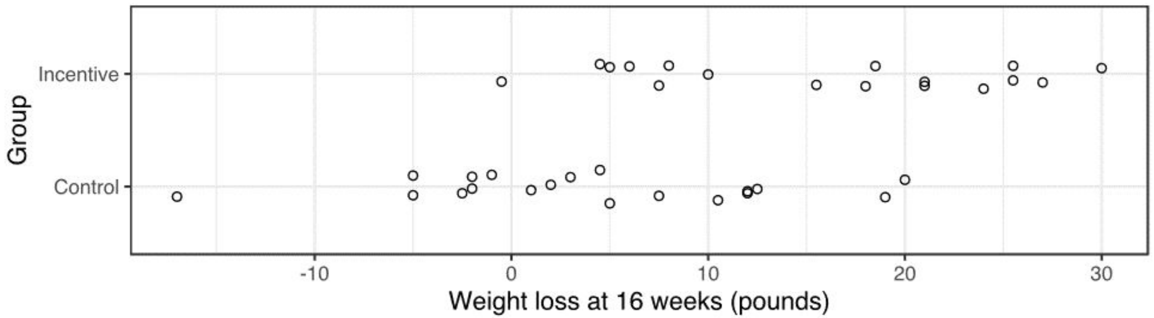
Participants are randomly assigned to a group

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

The group with incentives lost more weight on average than the group without incentives.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = 15.67

Mean_Control = 3.92

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = 11.75

Group 2

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

People in the VA medical center participating in the study

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: Treatment group

Type: Categorical

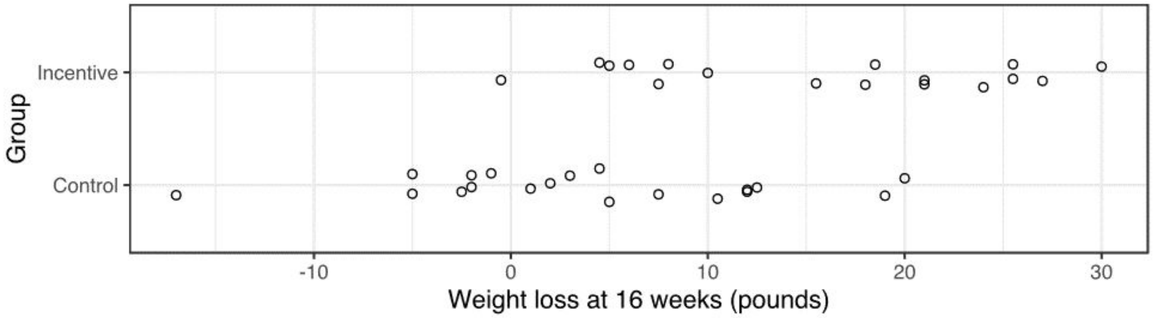
Response: The amount of weight loss

Type: Numerical

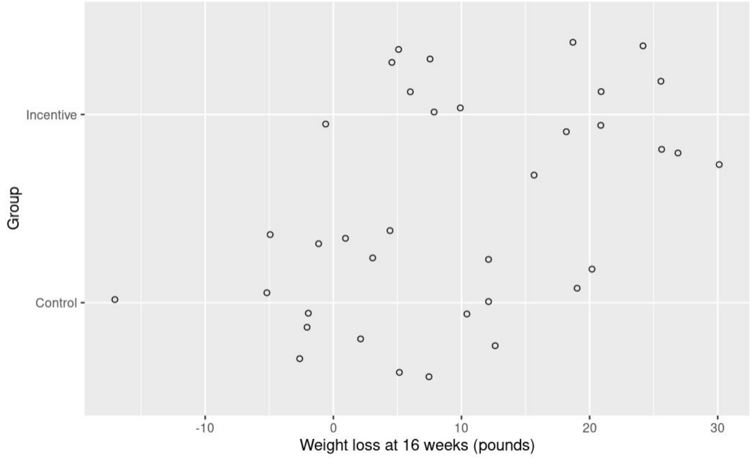
What makes this study an experiment? Explain briefly.

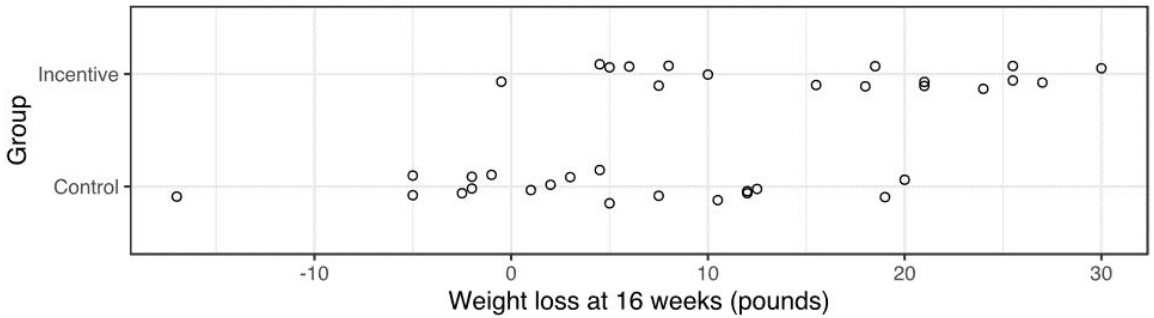
Participants are randomly assigned to one of three plans (control vs. treatment groups)

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

This graph shows the distribution of weight loss (lbs) after 16 weeks by treatment group. Just by looking at the graph, the typical weight loss of the incentive group is higher than those in the control group, even taking into account the outlier(s).

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = 15.676471

Mean_Control = 3.921053

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = 15.676471-3.921053 = 11.75542

Group 3

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

Healthy participants at the Philadelphia Veteran’s Affairs Medical Center

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory:

- Diet program

Type:

- Categorical

Response:

- Change in weight

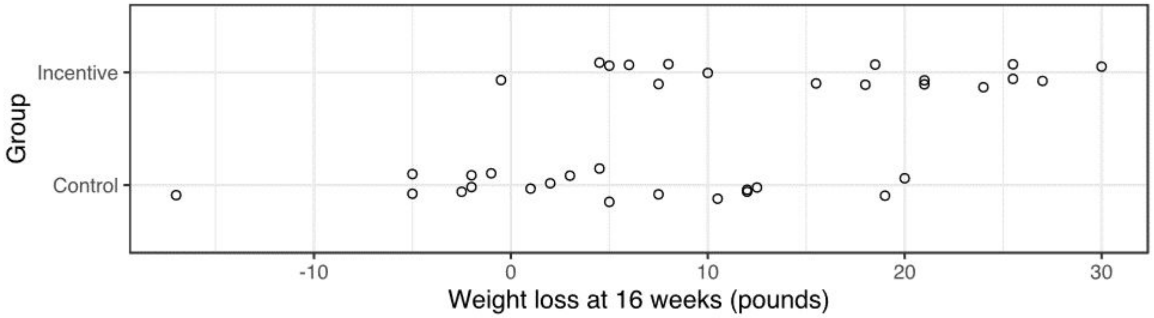
Type:

- Numerical

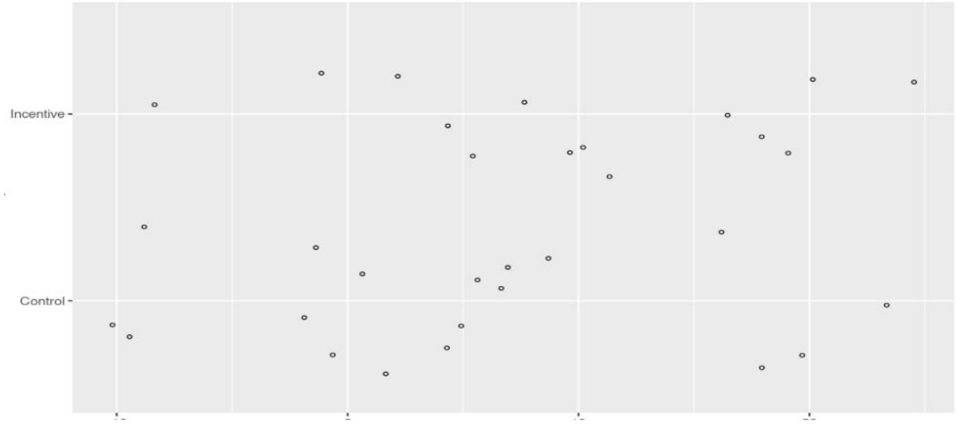
What makes this study an experiment? Explain briefly.

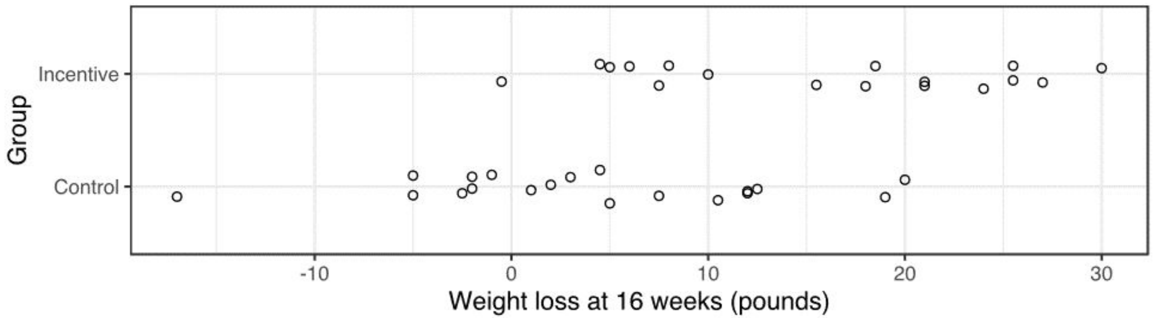
- It is a random study with control groups, each participant is randomly put into one of three groups, and the participants are randomly selected according to certain criteria.

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

Control mean- 3.92

Incentive mean- 15.676

Incentive appears to have lost more weight on average compared to the control group.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive =

Mean_Control =

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control =

Group 4

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

The healthy adults from the Philadelphia Veteran's Affairs Medical Center.

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: Financial incentive

Type: Categorical

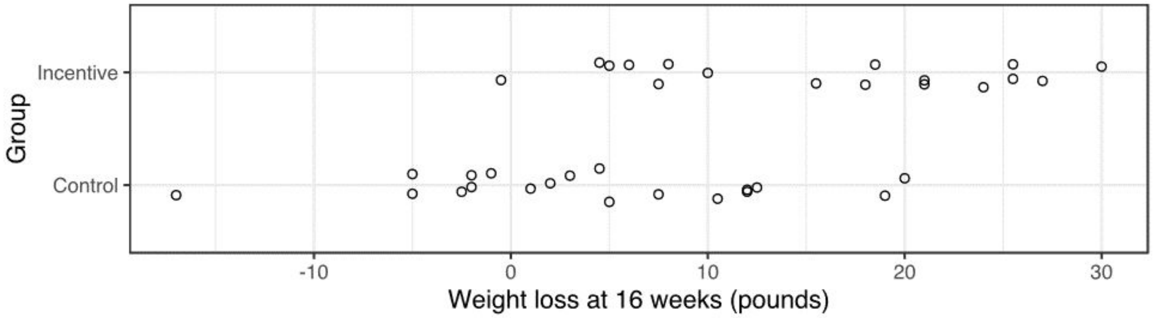
Response: Weight lost

Type: Numerical

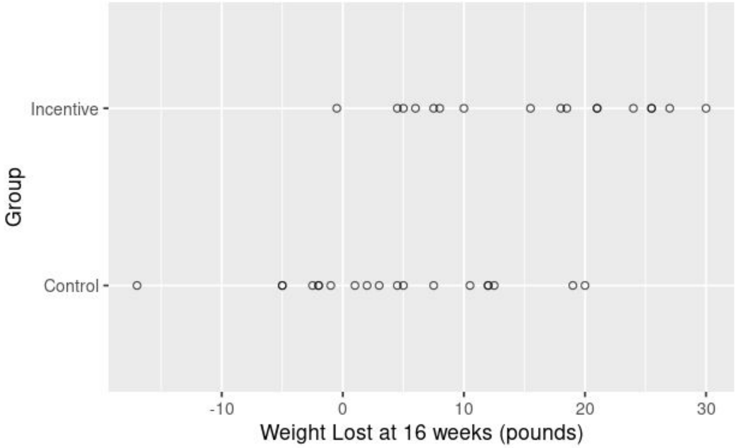
What makes this study an experiment? Explain briefly.

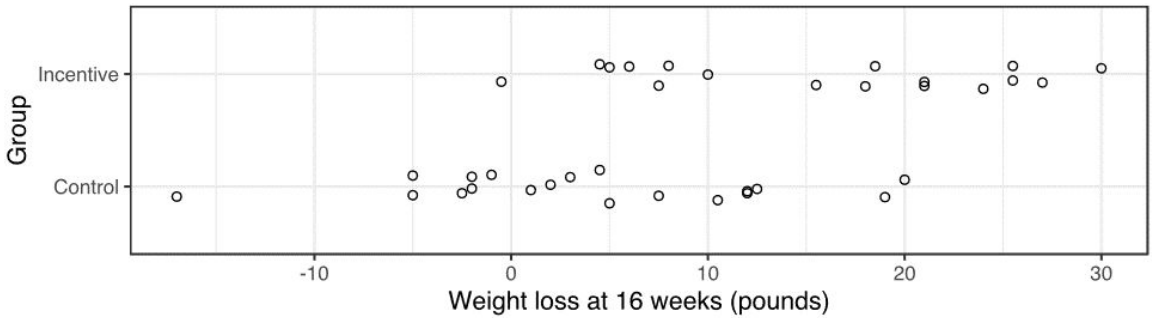
The researchers are controlling the variables and the participants are randomly assigned

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

The average person in the control group lost 3lbs, while the average person in the financial incentive group lost 18lbs. The middle 95% of people in the control group ranged from gaining 11.6lbs to losing 19.55lbs. The middle 95% of people in the financial incentive group lost between 1.5lbs and 28.8lbs.

The financial incentive seems to make a big impact on the amount of weight people lost, as people in the group receiving a financial incentive tended to lose more weight than people in the control group (without a financial incentive).

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

The coefficient 11.755 and the intercept 3.921 are useful for describing the data

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = 15.68

Mean_Control = 3.92

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = 11.76

Group 5

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

Participants were healthy adults, aged 30 to 70, with BMI between 30 and 40.

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: Weight loss plan

Type: Categorical

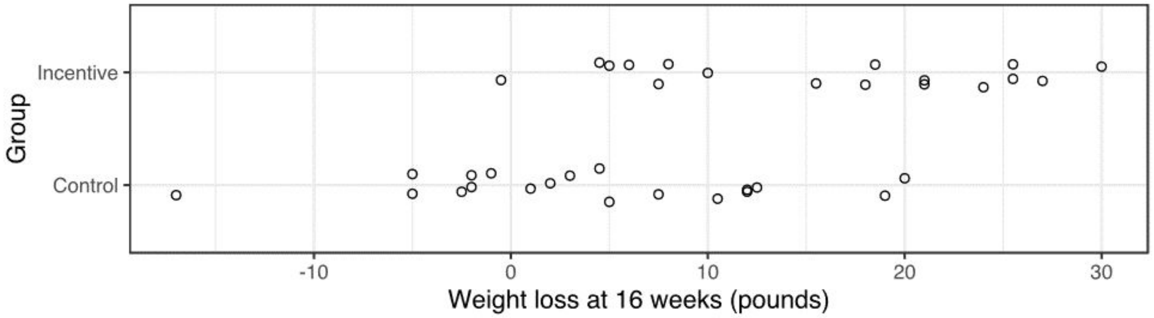
Response: Weight loss at 16 weeks

Type: Numerical

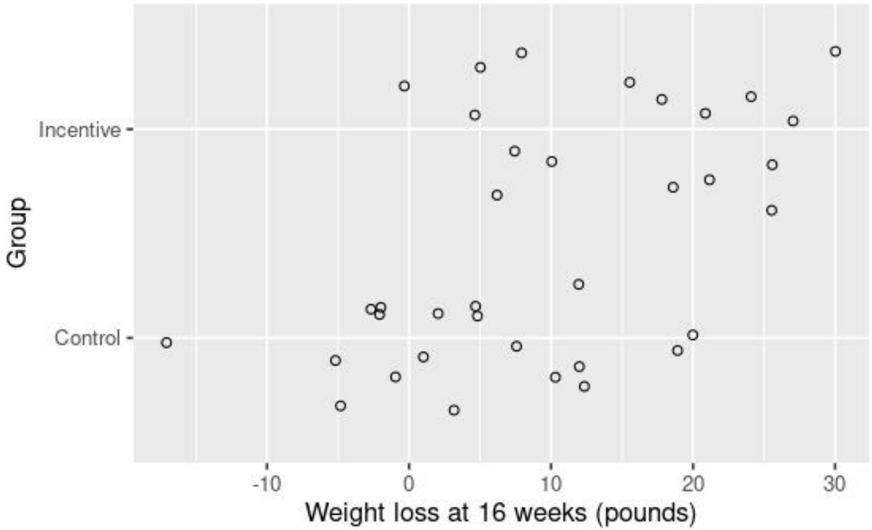
What makes this study an experiment? Explain briefly.

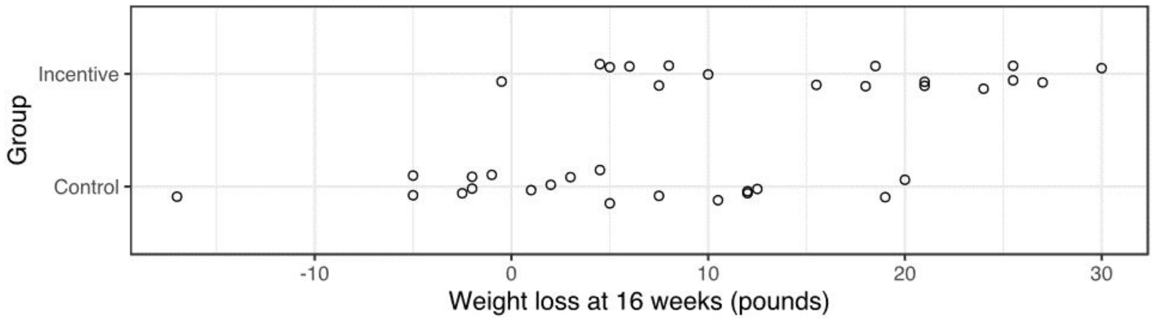
The randomly assigned weight-loss plans make the study an experiment since there is manipulation of a variable.

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

In general, the adults in the incentive group lost more weight at 16 weeks in comparison to the adults in the weekly weigh-in group. There is a larger amount of variability in the weekly weigh-in group, were weight loss at 16 weeks ranges from -5 to 20 pounds; in the incentive group weight loss results range from 0 to 30 pounds. Those in the weekly weigh-in group had more participants gain weight than that of the incentive group.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Record the mean weight loss after 16 weeks for each group: **9.472 lbs**

Mean_Incentive = **15.676 lbs**

Mean_Control = **3.921 lbs**

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = **11.755 lbs**

Group 6

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

Volunteers from the Philadelphia Veteran's Affairs Medical Center and were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40.

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: What type of Incentive Program were they on?

Type: Categorical

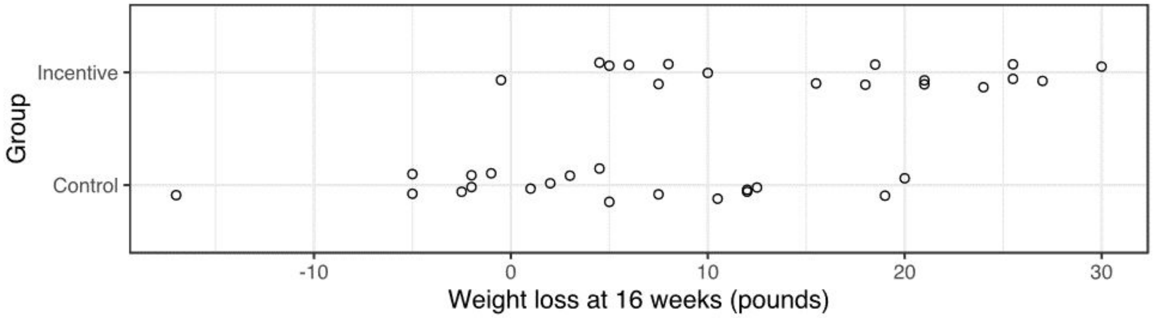
Response: How much weight did they lose?

Type: Numerical

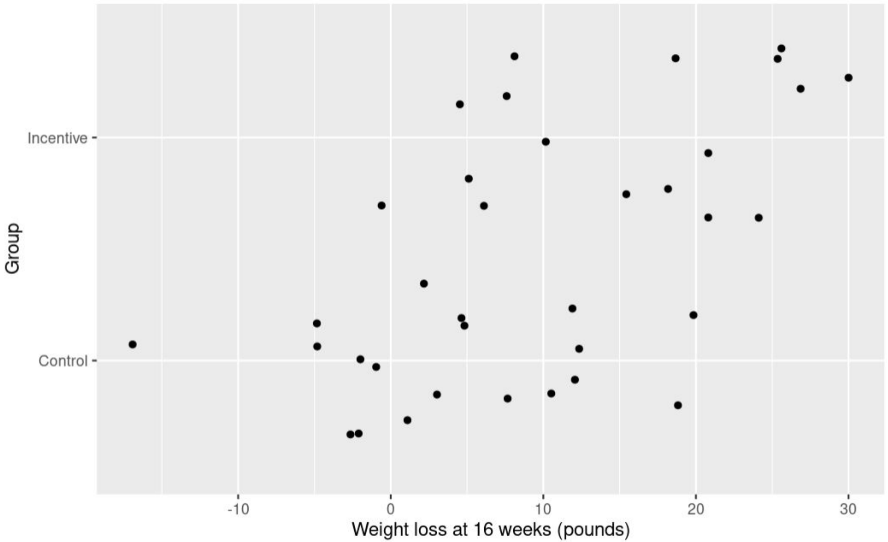
What makes this study an experiment? Explain briefly.

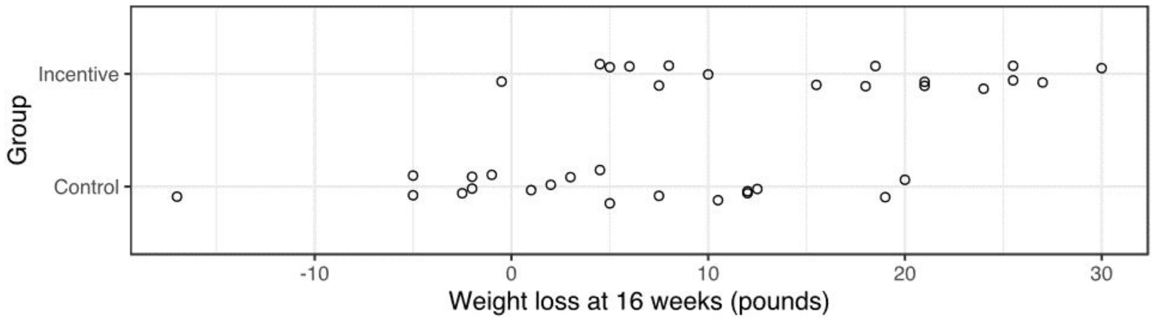
Control Group

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

Those with financial incentives were more likely to lose more weight on average than those in the control group.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don’t to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Those in the incentive group would lose typically 11.75lbs (slope) more than those in the control group with the typical control group individual losing roughly 3.92lbs.

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = **15.676471**

Mean_Control = **3.921053**

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = **11.755418**

Group 7

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

The recruited participants at the Philadelphia Veterans Affairs Medical Center

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: **Financial incentive**

Type: **Categorical**

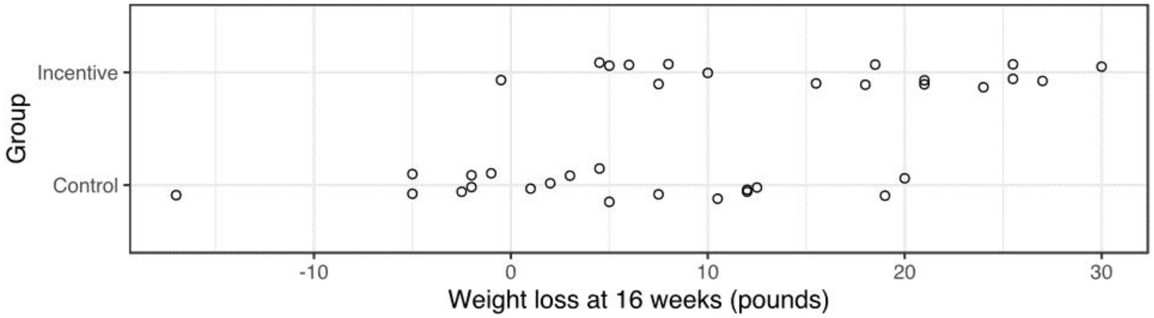
Response: **If participants lost weight**

Type:categorical

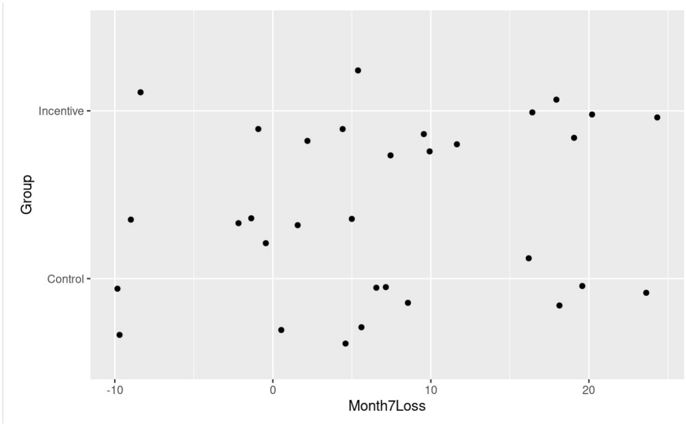
What makes this study an experiment? Explain briefly.

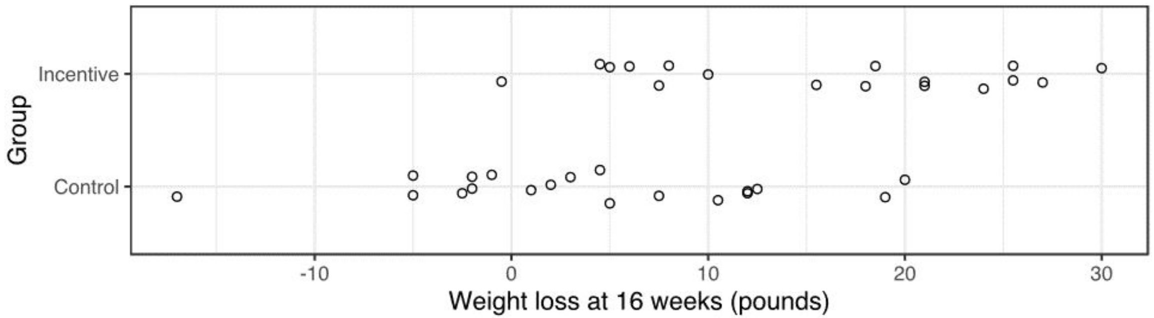
It’s an experimental study because the researchers were actively changing the explanatory variables by providing certain groups with financial incentive

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

Preliminary conclusions we can draw are that having an incentive works, because the average weight loss at 16 weeks is higher than the control group. The range for the incentive group is also smaller and shifted right compared to the control, indicating that the lowest amount of weight lost in the incentive group was higher than the control group.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

4.639: y-intercept. Describes the weight lost at 0 weeks
5.290: Slope. Describes the amount of weight lost per week

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = 9.928571

Mean_Control =

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control =

Group 8

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call “Deposit Contract.”

Who or what are the cases in the study?

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory:

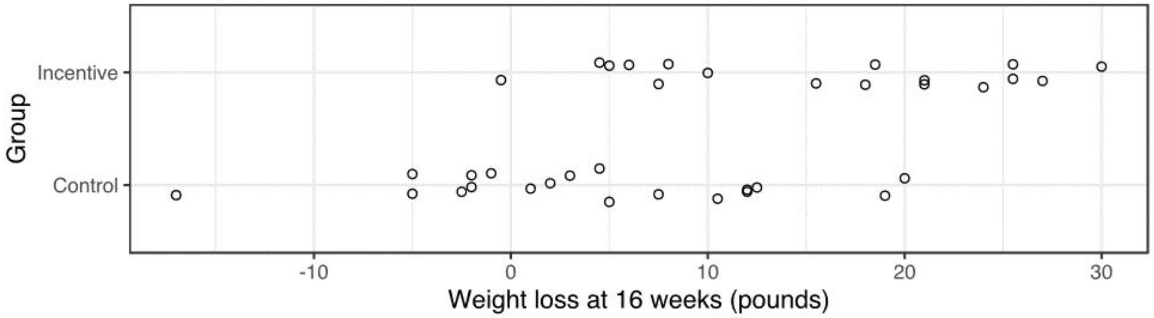
Type:

Response:

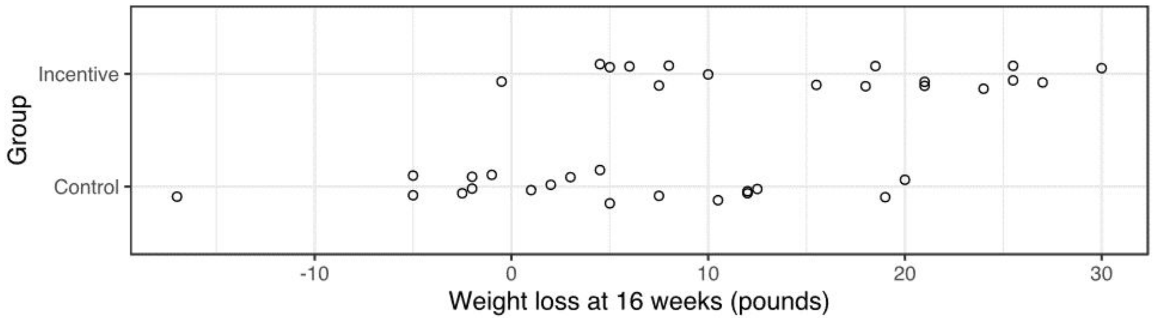
Type:

What makes this study an experiment? Explain briefly.

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.



Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive =

Mean_Control =

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control =

Group 9

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

The participants at the veterans medical center

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: Which weight loss plan

Type: categorical

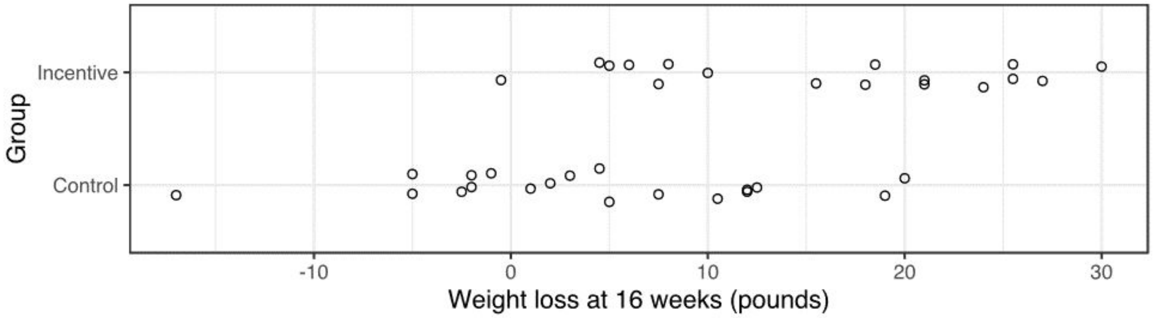
Response: weight loss

Type: numerical

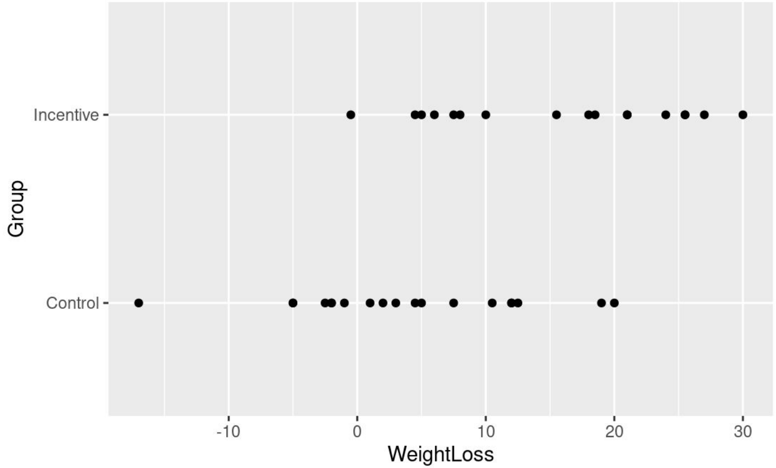
What makes this study an experiment? Explain briefly.

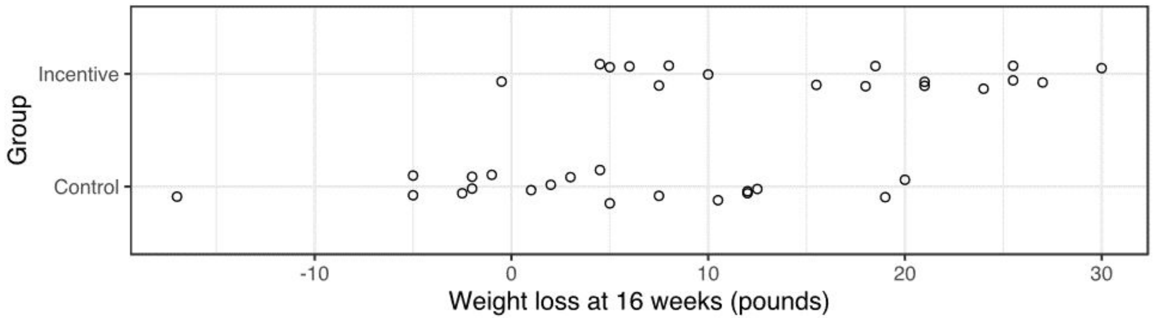
It's an experiment because this is being controlled

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

Overall, I would say that the incentive group seemed to favor spreading more on the losing weight side over the control group, with more points on the right side rather than the left

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)
As for what is important, the multiple r-squared is likely the best

Visual Outline

```
Residuals:
     Min       1Q   Median       3Q      Max
-20.9211  -6.7349   0.2012   8.0789  16.0789

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)       3.921      2.123   1.847 0.073451 .
GroupIncentive   11.755      3.089   3.805 0.000564 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.253 on 34 degrees of freedom
Multiple R-squared:  0.2987,	Adjusted R-squared:  0.2781
F-statistic: 14.48 on 1 and 34 DF,  p-value: 0.0005635
```

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = 15.676471

Mean_Control = 3.921053

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = 11.755418

Group 10

Researchers at the University of Pennsylvania conducted a study of an approach to weight loss inspired by behavioral economics in 2007-8. They randomly assigned participants in their study to one of three weight-loss plans. One weight-loss plan (the control) involved only weekly weigh-ins. The other two weight-loss plans both involved financial incentives, designed following the advice of behavioral economists, to encourage weight loss.

The researchers recruited participants at the Philadelphia Veteran's Affairs Medical Center in May through August 2007 and followed participants through June 2008. Participants were required to be healthy adults, aged 30 to 70, with BMI between 30 and 40. The authors described their sample as follows:

No significant differences were found in the baseline characteristics of any of the groups. The sample was predominantly male, with total household incomes of about $30,000, and a mean enrollment BMI of between 33.8 and 35.5 across the 3 groups. Participants in all 3 groups rated the importance of controlling their weight to be high on a 10-point scale (9.11-9.31) and had high confidence in their ability to lose weight (8.32-8.47).

Before starting on the plans, all of the participants had individual meetings with dieticians to discuss diet and exercise strategies for losing weight. The main part of the study gathered data focused on a period of 16 weeks, where the participants were given a goal of losing one pound per week. At the end of the 16 weeks, the total amount of weight loss was recorded for each participant. During the period of the study, participants in the control group only had weekly weight checks, but the two incentive plans provided different types of financial incentives for participants to meet their weekly weight-loss goals. We will focus on comparing the control plan with <u>one</u> of the incentive plans, which the authors call "Deposit Contract."

Who or what are the cases in the study?

The cases in the study are the male veterans at the hospital

What are the explanatory and response variables? Tell the type of each. (Categorical or numerical.)

Explanatory: The plan for the dieting

Type: categorical

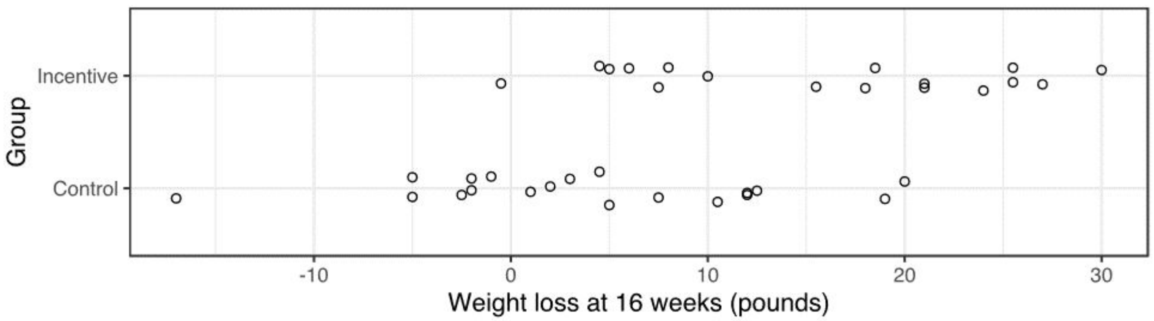
Response: the weight loss during the weeks

Type: numerical

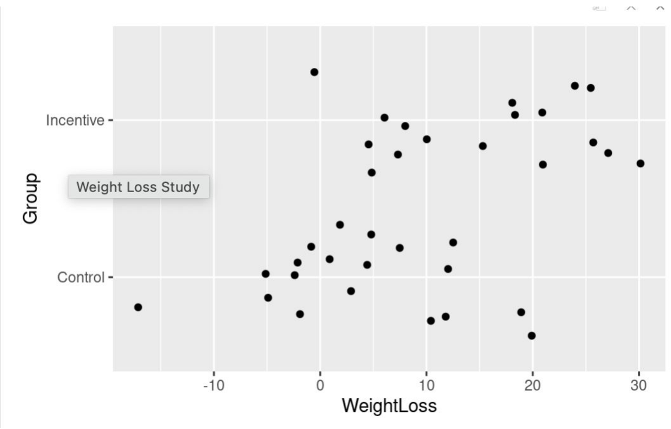
What makes this study an experiment? Explain briefly.

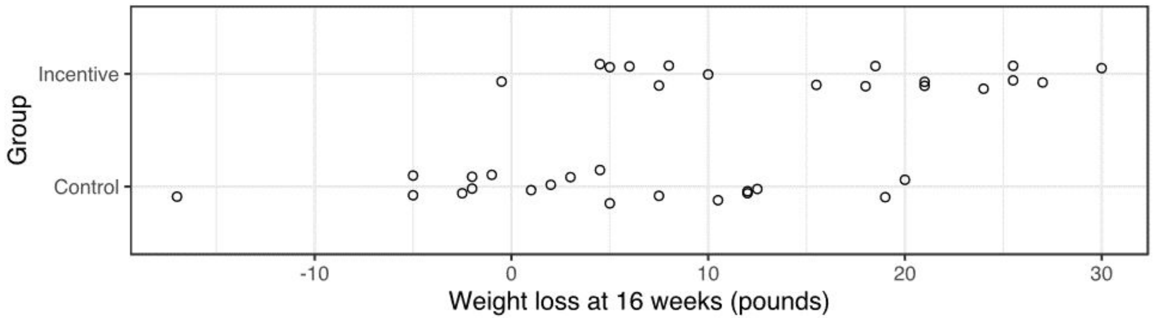
This study is a experiment because the experimenters are controlling a group by only doing weekly weigh ins and the other two with financial incentives

In a previous semester, Dr. Parson made this graph of these data:



In R, reproduce, as best you can, the graph above. Paste your version of the graph below.





Compare the distributions of weight loss in the two groups. Focus on the center and the variability of weight loss totals in each group. What preliminary conclusions do you draw from this summary of the study?

In the data the two groups are grouped on the graph differently. The control has seemed to not lose as much weight as the incentive group. The average weight loss for the incentive group was 18 pounds, while the control group was only 4 pounds.

Reference: The full article with details about this experiment is at https://jamanetwork.com/journals/jama/fullarticle/183047 . You don't to look at this now, but it may be helpful for us to have a link for later reference.

Follow the instructions in R to use the `lm` command to compare the two groups. Also print a summary of the model output. What numbers from the “summary” of your “lm” command are useful for describing the data? (Report numbers!)

```
Weight Loss Study
lm(formula = WeightLoss ~ Group, data = weight_loss)

Residuals:
     Min       1Q   Median       3Q      Max 
-20.9211  -6.7349   0.2012   8.0789  16.0789 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)       3.921      2.123   1.847 0.073451 .  
GroupIncentive   11.755      3.089   3.805 0.000564 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.253 on 34 degrees of freedom
Multiple R-squared:  0.2987,	Adjusted R-squared:  0.2781 
F-statistic: 14.48 on 1 and 34 DF,  p-value: 0.0005635
```

Record the mean weight loss after 16 weeks for each group:

Mean_Incentive = The mean for incentive was 15.6 pounds

Mean_Control = The mean for the control is 4 pounds

Compute the difference between mean weight loss for each group in this order:

Mean_Incentive - Mean_Control = 11.6